ControlShield: Network Security and Forensics for Industrial Control Systems

Group 7

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Project Objective

The objective of this project is to apply network forensics and cybersecurity techniques to a Programmable Logic Controller (PLC) and Industrial Control System (ICS) environment. PLCs are commonly used to control industrial processes and can be vulnerable to unauthorized network access, configuration changes, and malicious commands.

Our team will develop a network-based monitoring and security system, referred to as the **PLC Guardian**, that monitors network communication involving a PLC and analyzes the traffic for suspicious or unauthorized activity. The system will establish a baseline of normal PLC communication, identify authorized and unauthorized sources, detect relevant configuration or process changes, and maintain logs that can be used for forensic investigation.

We will use a Schneider Electric M221 PLC, an Ethernet switch, and Machine Expert Basic software to create a small industrial network environment. Network traffic will be captured during normal PLC operation, authorized changes, and simulated unauthorized activity. PCAP files will then be analyzed using Wireshark and our custom software.

The project will focus on four primary experiments:

1. **Normal PLC Traffic:** Capture and analyze normal communication to establish a baseline of expected network behavior.
2. **Authorized Change:** Perform a legitimate PLC configuration or process change and verify that the system detects, logs, and permits the activity.
3. **Unauthorized Change:** Simulate an unauthorized system attempting to communicate with or modify the PLC and verify that the system detects, logs, and blocks the activity where feasible.

4. **Forensic Investigation:** Analyze a mixed PCAP containing normal, authorized, and unauthorized activity and determine what occurred, which system was responsible, and whether the activity was authorized.

The project will combine network monitoring, packet analysis, cybersecurity, and digital forensics into a practical demonstration using an actual PLC environment.

Project Deliverables

1. Ten-Minute Demonstration Video

We will produce a **10-minute video demonstration** showing the design, implementation, and testing of the project.

The demonstration will include:

- Overview of the PLC and network architecture.
- Explanation of the PLC Guardian system design.
- Demonstration of normal PLC traffic.
- Demonstration of an authorized PLC change.
- Demonstration of an unauthorized communication or change attempt.
- Demonstration of detection, logging, and blocking.
- Demonstration of forensic analysis using a captured PCAP.
- Discussion of the project's results and limitations.

2. Source Code and Git Repository

We will provide the complete source code for the software developed as part of the project. Depending on the implementation requirements and the networking and packet-analysis libraries available, the PLC Guardian software will be developed in **C or Python**.

The project will be maintained in a **Git repository** throughout development. The repository will contain the source code, documentation, configuration files, testing materials, and other relevant project files. Using Git will allow the team to collaborate on development, track changes, and maintain different versions of the project.

The source code will include functionality for:

- Processing and analyzing network traffic.
- Identifying relevant PLC communication.
- Identifying authorized and unauthorized sources.
- Detecting suspicious activity.
- Recording security and forensic events.
- Supporting the blocking of unauthorized communication where feasible.
- Processing or assisting with the analysis of PCAP files.

Wireshark and PCAP files will also be used as supporting tools for packet capture and network forensic analysis.

3. Technical Report

We will provide a technical report documenting the complete project according to the course guidelines.

The report will cover:

- **Project Background and Motivation**
    - PLCs and Industrial Control Systems
    - Security risks associated with network-connected PLCs
    - Importance of network forensics in ICS environments
- **System Design**
    - PLC and network architecture
    - Network monitoring architecture
    - PLC Guardian design
    - Authorized-source identification
    - Detection and logging mechanisms
    - Blocking approach
- **Development and Implementation**
    - Software architecture
    - C or Python implementation
    - Packet capture and PCAP processing
    - Integration with network forensic tools
    - Logging and event-recording mechanisms
- **Testing**
    - Normal PLC traffic experiment
    - Authorized change experiment
    - Unauthorized change experiment
    - Forensic investigation experiment
- **Results and Analysis**
    - Detection results
    - Logging results
    - Blocking results
    - Forensic reconstruction of events
    - False positives and false negatives
    - System limitations

- **Running Procedures**
    - PLC and network setup
    - Software installation and configuration
    - How to capture traffic
    - How to run the PLC Guardian
    - How to analyze captured traffic
    - How to reproduce the experiments
- **Conclusion and Future Work**
    - Summary of project results
    - Effectiveness of the proposed system
    - Potential improvements and real-world applications

Project Summary

The project will use an available Schneider Electric M221 PLC with an Ethernet communication interface, a Netgear Ethernet switch, and Machine Expert Basic software. This hardware will allow the team to generate realistic PLC network traffic rather than relying entirely on simulated data.

Network traffic will be captured as PCAP files and analyzed using Wireshark and the team's custom software. The development environment will support both Windows and Linux where practical, allowing the project to incorporate cross-platform networking and forensic analysis.

The final project will demonstrate an end-to-end workflow in which network traffic is **captured, monitored, analyzed, classified, logged, and, when appropriate, blocked**, while also providing sufficient evidence for a subsequent forensic investigation.